

Integration Contracts

Every external system that plugs into the ISR C2 platform speaks against one of the contracts in this book. Sensors, NN classifiers, receiver agencies, dispatch APIs, IAM providers, agentic services, audit sinks — each has a documented shape any implementer can build against without asking us.

Read this before writing any adapter, integration, or customer-facing endpoint.

Version	Date	Source of truth
0.1 (Phase A initial)	2026-09-13	Working tree at HEAD

Structure

Every contract section follows the same shape:

1. **Purpose** — what integrates through this contract
2. **Actors** — who calls, who receives
3. **Wire format** — concrete schemas
4. **Auth model**
5. **Error semantics**
6. **Sequence diagram** — mermaid
7. **Example** — request / response
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Cross-cutting conventions

Applies to every contract in this book.

- **Encoding.** UTF-8 throughout. Never Latin-1. Danish characters (æ, ø, å) always preserved.
- **Time.** ISO 8601 with explicit `Z` suffix (UTC). Sub-second precision at millisecond resolution when ordering matters.
- **Coordinates.** WGS 84 latitude/longitude in decimal degrees. Altitude in metres above ellipsoid (WGS 84), NOT above ground, NOT above sea level. All calculations round-trip through Cesium's Cartographic type.

- **Identifiers.** Kebab-case strings for machine ids (`shahed-136` , `politi-kbh` , `evt-1234`). Freeform strings for human labels. Never mix.
- **Auth.** Every contract uses OAuth 2.0 client-credentials against the Auth+IAM contract (Section 12) unless explicitly noted. Tokens carry tenant + role claims.
- **Idempotency.** Every write endpoint accepts an `Idempotency-Key` header. Repeat requests with the same key + same payload return the original response. Repeat with same key + different payload returns HTTP 409.
- **Error format.** All 4xx / 5xx responses use RFC 7807 Problem Details JSON:

```
{
  "type": "https://isr-systems.dk/errors/invalid-signature",
  "title": "Signature payload rejected",
  "status": 400,
  "detail": "acoustic.confidence must be in [0.0, 1.0]",
  "instance": "/sensors/cph-radar-01/detections",
  "correlation-id": "abc-123"
}
```

- **Correlation.** Every request must carry an `X-Correlation-ID` header (UUID v4). Propagates through every downstream call. Surfaces in audit log + feedback log.
- **Rate limits.** Per-tenant, per-endpoint. Enforced at the API layer. `429 Too Many Requests` includes `Retry-After` header in seconds.
- **Versioning.** Every endpoint is versioned in the path: `/v1/sensors/...` . Breaking changes bump the major version + run parallel for one deprecation cycle.
- **Detection-only.** No contract in this book triggers a real-world dispatch action without a human confirming through the Mission Console. Every dispatch adapter (Section 6) requires a signed operator confirmation payload — the raw asset-command endpoint doesn't exist.

1. Sensor ingestion

Purpose

Every physical sensor (RF spectrum analyser, acoustic array, EO/IR camera, radar) posts detections into the platform through this contract. This is the entry point for real-world signal.

Actors

- **Caller:** sensor device or edge gateway on the customer's site
- **Receiver:** ISR C2 sensor ingest API (Azure Function endpoint per site tenant)

Wire format

POST `/v1/sensors/{sensor_id}/detections`

Batch endpoint. Sensors send one array of detections per second (or per fixed window). Batching amortises TLS handshake + reduces per-detection overhead.

Request body:

```
{
  "sensor_id": "cph-rf-01",
  "batch_id": "batch-1726220000-001",
  "window_start": "2026-09-13T10:30:00.000Z",
  "window_end": "2026-09-13T10:30:01.000Z",
  "detections": [
```

```

{
  "detection_id": "det-1726220000-abc",
  "timestamp": "2026-09-13T10:30:00.412Z",
  "modality": "rf",
  "position": {
    "lat": 55.617,
    "lon": 12.646,
    "alt_m": 120.5,
    "accuracy_m": 15.0
  },
  "signature": {
    "rf": {
      "band": "2.4-GHz",
      "center_hz": 2437000000,
      "bandwidth_hz": 20000000,
      "power_dbm": -68.2,
      "modulation_hint": "ofdm",
      "hopping_pattern": null
    }
  },
  "confidence": 0.87,
  "raw_reference": "s3://cph-rf-01-raw/2026-09-13/10-30-00-412.iq"
},
{
  "detection_id": "det-1726220000-def",
  "timestamp": "2026-09-13T10:30:00.520Z",
  "modality": "acoustic",
  "position": { "lat": 55.618, "lon": 12.647, "alt_m": null, "accuracy_m": 30.0 },
  "signature": {
    "acoustic": {
      "profile": "moped-buzz",
      "fundamental_hz": 78.0,
      "harmonic_ratio": 0.42,
      "spl_db": 54.3
    }
  },
  "confidence": 0.91,
  "raw_reference": "s3://cph-acoustic-01-raw/2026-09-13/10-30-00-520.wav"
}
]
}

```

Response 202 Accepted:

```

{
  "batch_id": "batch-1726220000-001",
  "accepted_count": 2,
  "rejected_count": 0,
  "rejections": [],
  "next_batch_after": "2026-09-13T10:30:01.000Z"
}

```

Field constraints:

- `sensor_id` — kebab-case, must match a registered sensor in the site definition (Section 3).

- `modality` — one of `rf`, `acoustic`, `visual`, `radar`, `cooperative-traffic` (Section 11). Determines which `signature.*` sub-object is required.
- `position.accuracy_m` — 1-sigma horizontal error estimate. Used by the fusion layer for weighting.
- `confidence` — sensor-side belief this detection is a real object (not noise). Range [0.0, 1.0].
- `raw_reference` — optional pointer to the raw sample in Blob storage. Enables replay + forensic audit. Sensor is responsible for uploading the raw file BEFORE posting the detection metadata.
- `signature.{modality}` — modality-specific schema. See per-modality sub-sections below.

Signature sub-schemas

RF signature:

Field	Type	Notes
band	enum	Values from <code>threat_taxonomy.js</code> <code>RF_BANDS</code> (2.4-GHz, 5.8-GHz, 900-MHz, L-band, S-band, X-band, Ku-band, Ka-band, HF, VHF, UHF, silent)
center_hz	number	Center frequency in Hz
bandwidth_hz	number	Occupied bandwidth in Hz
power_dbm	number	Received signal power in dBm
modulation_hint	enum null	Best guess: <code>ofdm</code> , <code>fsk</code> , <code>qpsk</code> , <code>unknown</code> , or null when not determinable
hopping_pattern	object null	If frequency-hopping detected: { <code>pattern_hash</code> , <code>dwell_ms</code> , <code>hop_count</code> }

Acoustic signature:

Field	Type	Notes
profile	enum	Values from <code>threat_taxonomy.js</code> <code>ACOUSTIC</code> (moped-buzz, high-whine, buzz, prop-hum, piston-drone, turbine-roar, jet-scream, rotor-thump, tilt-rotor, swarm-chorus, silent-electric, unknown)
fundamental_hz	number	Detected fundamental frequency
harmonic_ratio	number [0-1]	Strength of harmonic content vs noise floor
spl_db	number	Sound pressure level in dB

Visual signature:

Field	Type	Notes
silhouette	enum	Values from <code>threat_taxonomy.js</code> <code>VISUAL</code>
bbox_px	[x, y, w, h]	Detection bounding box in source image
image_reference	string	Blob URL pointing at the cropped detection thumbnail
range_m	number	Estimated range to target

Radar signature:

Field	Type	Notes
-------	------	-------

rsc_m2	number	Radar cross-section in square metres
doppler_ms	number	Radial velocity in m/s
track_id	string	Radar's own internal track identifier (used for de-dupe across batches)

Auth model

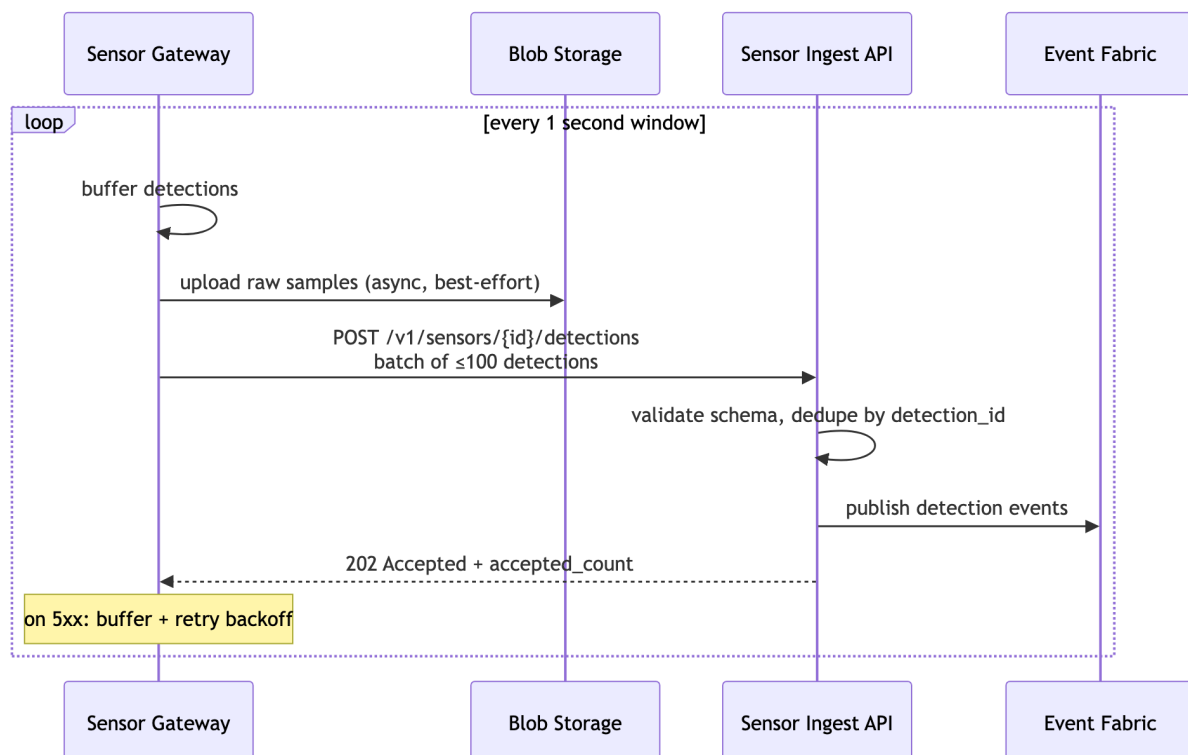
- OAuth 2.0 client credentials. Sensor gateway registers as a client with `sensor:write` scope + a tenant claim.
- Every request signed with `Bearer <access_token>` header.
- Access token TTL: 1 hour. Gateway refreshes 5 min before expiry.

Error semantics

- **400 Bad Request** — malformed payload, missing required field, coordinate out of range. Response `type` identifies which field failed.
- **401 Unauthorized** — token expired, invalid, or revoked.
- **403 Forbidden** — `sensor_id` not owned by the calling tenant.
- **409 Conflict** — `batch_id` already ingested (idempotency hit). Original response is echoed.
- **413 Payload Too Large** — batch exceeds 1 MB. Split and retry.
- **429 Too Many Requests** — rate limit. Retry after `Retry-After` seconds.
- **503 Service Unavailable** — server-side backpressure. Retry with exponential backoff (max 5 attempts).

Sensor gateway MUST NOT drop detections on 5xx. Buffer locally and retry until 2xx or the buffer's TTL expires.

Sequence diagram



Example

Curl request:

```
curl -X POST https://api.isr-systems.dk/v1/sensors/cph-rf-01/detections \
-H "Authorization: Bearer $TOKEN" \
-H "Content-Type: application/json" \
-H "X-Correlation-ID: 550e8400-e29b-41d4-a716-446655440000" \
-H "Idempotency-Key: batch-1726220000-001" \
-d @batch.json
```

Reference implementation

- **Mock ingest endpoint:** `scripts/dev/mock-sensor-ingest.mjs` (stub — accepts any payload, logs to stdout, returns 202)
- **Sensor gateway reference:** to be published as a separate repo (`isr-sensor-gateway`) once wire format stabilises
- **Sample batches for testing:** `scripts/dev/sample-detections/*.json`

Open questions

- **Multi-modality per detection:** should a single detection carry both RF and acoustic signature slices when the sensor is a fused array? Current design says no (post as two detections, let fusion layer correlate). Revisit if bandwidth becomes an issue.
- **Backpressure signal:** currently 503 + Retry-After. Consider adding WebSocket push for immediate backpressure indication when API is under load.
- **Raw sample upload timing:** async best-effort today. Should we require raw upload confirmation before accepting detection metadata? Trade-off: latency vs forensic completeness.

2. NN classifier

Purpose

Raw detections from the sensor layer (Section 1) need to be classified — is this a DJI Mavic, a Shahed-136, or noise? The NN classifier contract specifies how a machine-learning service (running as an Azure ML endpoint or a self-hosted inference server) receives batched detections and emits classified events.

Actors

- **Caller:** ISR C2 event fabric (server-side, per-tenant instance)
- **Receiver:** NN classifier service (Azure ML endpoint, or self-hosted GPU node)

Wire format

POST `/v1/classify`

Request body:

```
{
  "classify_id": "cls-1726220000-abc",
  "context": {
    "site_id": "cph",
    "site_domain_scope": ["aviation", "ground"],
    "timestamp": "2026-09-13T10:30:00.520Z"
  },
  "detections": [
    {
      "detection_id": "det-1726220000-abc",
```

```

    "modality": "rf",
    "signature": {
      "rf": {
        "band": "2.4-GHz",
        "center_hz": 2437000000,
        "bandwidth_hz": 20000000,
        "power_dbm": -68.2,
        "modulation_hint": "ofdm"
      }
    },
    "position": { "lat": 55.617, "lon": 12.646, "alt_m": 120.5 }
  },
  {
    "detection_id": "det-1726220000-def",
    "modality": "acoustic",
    "signature": { "acoustic": { "profile": "moped-buzz", "fundamental_hz": 78.0, "spl_db": 54.3
} } },
    "position": { "lat": 55.618, "lon": 12.647, "alt_m": null }
  }
]
}

```

Response 200 OK:

```

{
  "classify_id": "cls-1726220000-abc",
  "model_version": "isr-classifier-v0.3.2",
  "latency_ms": 87,
  "classifications": [
    {
      "detection_id": "det-1726220000-abc",
      "family": "commercial-quadcopter",
      "family_confidence": 0.82,
      "candidate_models": [
        { "id": "dji-mavic-3", "score": 0.51 },
        { "id": "dji-mavic-3-pro", "score": 0.28 },
        { "id": "dji-air-2s", "score": 0.11 }
      ],
      "classification": "unknown",
      "threat_estimate": "low",
      "rationale": "2.4-GHz + 5.8-GHz OFDM pattern consistent with DJI OcuSync 2.0 link. Model discriminator is inconclusive between Mavic 3 variants."
    },
    {
      "detection_id": "det-1726220000-def",
      "family": "loitering-munition",
      "family_confidence": 0.94,
      "candidate_models": [
        { "id": "shahed-136", "score": 0.71 },
        { "id": "shahed-131", "score": 0.19 }
      ],
      "classification": "hostile",
      "threat_estimate": "critical",
      "rationale": "Moped-buzz acoustic signature at 78Hz fundamental is a Shahed-class rotary engine tell. No cooperative flight plan for this position."
    }
  ]
}

```

```
]
}
```

Field constraints:

- `family` — MUST be a valid enum value from `threat_taxonomy.js` `FAMILIES` . Reject if the classifier can't map to a known family — return `unknown-signature` and let the routing matrix handle it.
- `family_confidence` — [0.0, 1.0]. Below 0.5 the routing matrix treats the family as `unknown-signature` regardless of the top pick.
- `candidate_models` — ordered by score descending. Model ids MUST exist in `threat_taxonomy.js` `MODELS` . Send at most 5 candidates.
- `classification` — enum: `hostile` , `friendly` , `unknown` , `resolved` . Classifier's initial guess; operator can override.
- `threat_estimate` — enum: `critical` , `high` , `medium` , `low` , `unknown` . Feeds the routing matrix's threat axis.
- `rationale` — one-line human-readable explanation. Surfaces in Agent B narrative + case-file audit.
- `model_version` — classifier's own version tag. Recorded on every classified event for later model-drift diagnosis.

Auth model

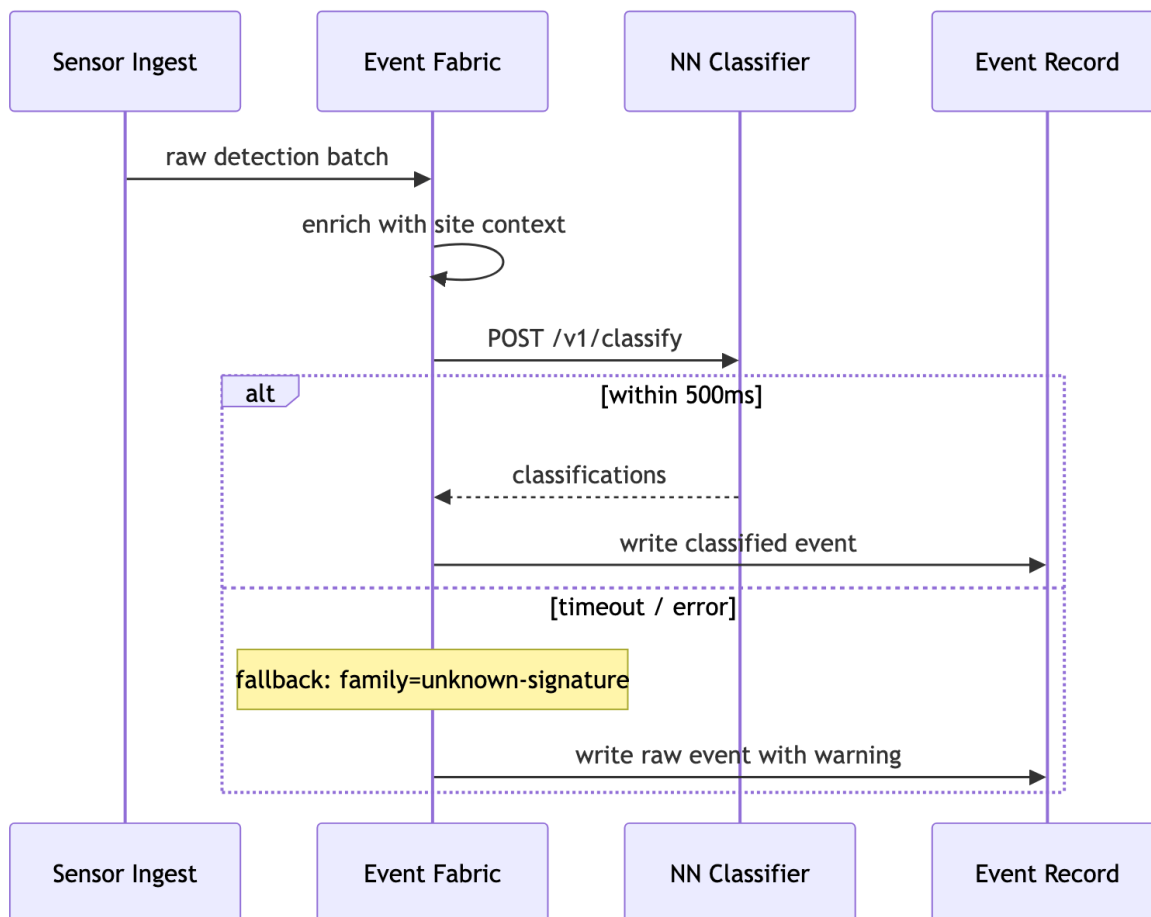
- Bidirectional mTLS between event fabric and classifier service.
- Classifier presents client cert; event fabric validates against a per-classifier trust anchor.
- Rotate certs every 90 days via Azure Key Vault.

Error semantics

- **400 Bad Request** — batch schema invalid.
- **408 Request Timeout** — classifier processing exceeded latency budget (default 500ms per batch). Event fabric proceeds with raw signature (family = `unknown-signature`).
- **422 Unprocessable Entity** — classifier produced output that fails the schema constraints above (e.g. invalid family value). Event fabric logs + falls back to `unknown-signature` .
- **500-503** — event fabric retries once with exponential backoff (100ms, 500ms). If both fail, detection proceeds with `unknown-signature` and a warning is written to the operator's Mission Console.

Latency budget: 500ms end-to-end per batch. Classifier services that can't meet this MUST be scaled horizontally, not slowed down. The event fabric will not wait past 500ms.

Sequence diagram



Example

Detection classified as Shahed → routing matrix (Section 12 of cross-agency-flows.md) fires strategic-strike + hostile-intel rules → auto-observers loop in.

Reference implementation

- **Mock classifier:** `scripts/dev/mock-nn-classifier.mjs` — returns deterministic mappings for known signature slices (moped-buzz → shahed-136, high-whine + 2.4GHz → commercial-quadcopter, etc). Useful for local testing without a real model.
- **Model versioning:** every event record stores `event.classifier_version` for later drift analysis.

Open questions

- **Batching vs streaming:** current design is request/response per batch. Consider WebSocket streaming for lower-latency use cases (e.g. tracking a fast-moving target across many frames).
- **Multi-modal classification:** should the classifier accept a single detection with multiple signature slices fused, or classify per-modality and let the platform reconcile? Current design: classifier gets per-detection input; platform-side fusion happens upstream in Section 11 (cooperative traffic fusion).
- **Explainability:** `rationale` is one line today. Should we add structured feature importance (which signature elements drove the classification)? Useful for regulator conversations.

3. Site definition

Purpose

Every customer site (airport, port, energy substation, data centre, government facility) is defined by a single YAML/JSON manifest that specifies boundaries, sensor placements, domain scope, tenant assignment, and default cascade recipients. The "7-step config-only add" claim rides on this schema.

Actors

- **Author:** ISR ops team (admin tenant) OR customer's own IT if they self-onboard
- **Consumer:** platform boot loader — reads all site manifests, registers each with `registerSiteDomains()` and the sensor + destination registries

Schema

`sites/{site_id}.yaml`

```
# Site identifier – kebab-case, unique across all tenants.
# Matches `event.siteId` throughout the codebase.
site_id: cph

# Human display label. Danish characters preserved.
label: Copenhagen Airport (Københavns Lufthavn)

# ICAO / IATA / port code – whatever the customer's operators call this
# site day-to-day. Optional but recommended for cross-reference.
code: CPH

# Tenant that owns this site. One of the OPERATORS in src/roles.js.
tenant: op-cph-airports

# Site type is DERIVED from domain_scope; kept as a hint for
# routing rules and doc generators.
site_type: airport

# Center point + boundary. Cesium consumes these for map rendering.
coordinates:
  lat: 55.618
  lon: 12.656

# Outer boundary as a closed GeoJSON-style polygon (lon, lat pairs).
# Coordinates in WGS 84 decimal degrees.
boundary:
  - [12.6301, 55.6284]
  - [12.6512, 55.6291]
  - [12.6598, 55.6202]
  - [12.6420, 55.6114]
  - [12.6303, 55.6183]
  - [12.6301, 55.6284]    # close the ring

# Which operational domains apply. Feeds event.domainScope and the
# threat_routing baseline rules. See DOMAIN enum in threat_routing.js.
domain_scope:
  - aviation
  - ground
```

```

# Sensors physically installed at this site. Each sensor is registered
# with the sensor ingestion contract (Section 1) at boot.
sensors:
  - sensor_id: cph-rf-01
    modality: rf
    position: { lat: 55.6181, lon: 12.6560, alt_m: 15 }
    coverage_radius_m: 3000
    label: "RF Spectrum Analyser (Terminal 3 rooftop)"
  - sensor_id: cph-acoustic-01
    modality: acoustic
    position: { lat: 55.6175, lon: 12.6540, alt_m: 8 }
    coverage_radius_m: 1200
    label: "Acoustic Array (Runway 22L)"
  - sensor_id: cph-radar-01
    modality: radar
    position: { lat: 55.6190, lon: 12.6600, alt_m: 45 }
    coverage_radius_m: 8000
    label: "Perimeter Radar (Control Tower)"
  - sensor_id: cph-eo-01
    modality: visual
    position: { lat: 55.6183, lon: 12.6555, alt_m: 25 }
    coverage_radius_m: 1500
    label: "EO/IR Camera (Terminal 3 pan-tilt-zoom)"

# Default cascade recipients when a hostile event occurs at this site.
# These are SUGGESTIONS pre-populated in the Mission Console; operator
# can accept, decline, or add more. Consumed by cascade_picker.js.
default_cascade_recipients:
  hostile_any:
    - pet
    - fe
    - politi-kbh
  hostile_high:
    - forsvarskmd
    - rigspoliti
    - beredskab
    - agency-traf

# Auto-observer role set that ALWAYS loops in on any event at this site,
# regardless of routing matrix. Empty by default; use sparingly.
always_observers:
  - agency-traf # Trafikstyrelsen is baseline for aviation domain anyway,
                # but naming here makes it explicit + audit-visible.

# Operating mode: 'live' (real sensors, operator-grade UX), 'sim'
# (cinematic drone POV for demos + training), or 'mixed' (per-detection
# source tag decides). Feeds the mode terminology per project memory.
mode: live

# Operator's dispatch scope – which asset kinds the operator's own team
# can dispatch. Empty array = SaaS-only tenant (no on-site assets).
# Referenced by _ROLE_DISPATCH_SCOPE_LOOKUP in main.js.
operator_dispatch_scope:
  - wildlife-response

# Airspace / regulatory context. Free-text notes surfaced in the

```

```
# case-file emphasis + Agent B narrative.
regulatory_context: |
  Class D controlled airspace up to 4500 ft AGL. Coordination with
  ATC on 118.100 MHz. NOTAM authority: Trafikstyrelsen. Standard
  drone incursion protocol: hold departures, delay arrivals >5 min,
  Trafikstyrelsen issues restriction NOTAM.

# Site-specific escalation SLA overrides (minutes). Defaults live in
# the escalation config; per-site overrides land here for sensitive
# sites. Empty means use platform defaults.
escalation_sla_minutes:
  hostile_high: 3      # override – tighter than default 5 min

# Reference to the per-site flow document that walks through the
# detection-to-closed-report lifecycle for this specific site.
flow_doc: docs/sites/cph-airport.md
```

Onboarding checklist

Adding a new site is a config-only change. The 7 steps:

1. **Create** `sites/{site_id}.yaml` using this schema.
2. **Add the tenant** to `src/roles.js` OPERATORS if it's a new customer (existing tenant: skip).
3. **Copy the site flow doc template** from `docs/sites/_template.md` to `docs/sites/{site_id}.md` and fill in.
4. **Register the sensors** in the sensor gateway config (per Section 1) so they know where to POST.
5. **Configure receiver adapter destinations** in `src/dispatch_source.js` if the customer has bespoke receiver APIs (per Section 5). Skip for SaaS-only tenants.
6. **Add site-specific test scenarios** to `scripts/dev/scenarios/` so simulation + regression tests cover the new site.
7. **Deploy**. The platform picks up the new site at next boot; no code changes.

Zero source files touched beyond the config manifest and (optionally) a receiver adapter registration for the new tenant.

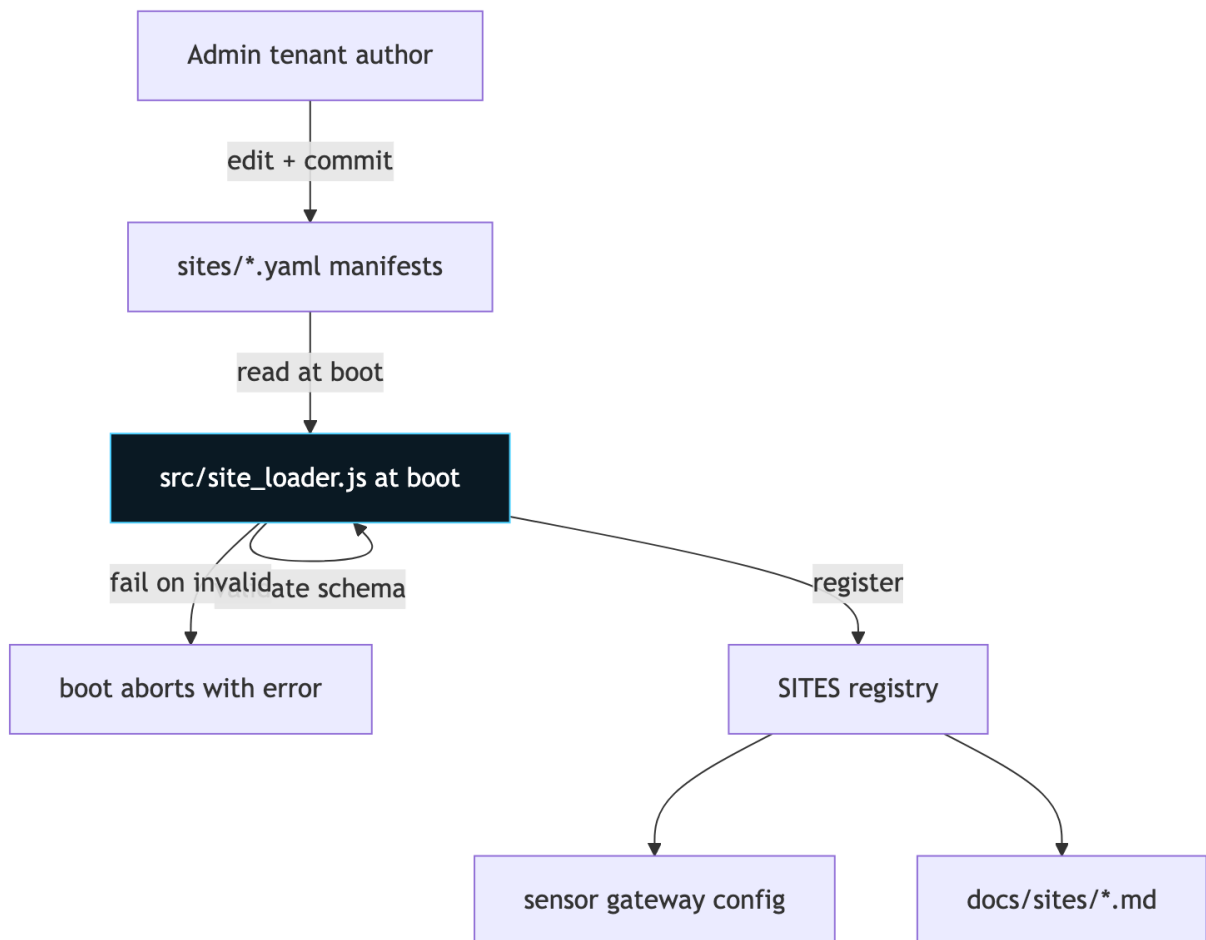
Validation

Every manifest is validated at boot by `src/site_loader.js` against a JSON Schema. Validation failures block boot with a clear error pointing at the manifest field + line number. Never ships a broken site to production.

Auth model

- **Read (any authenticated user)**: site metadata is public within a tenant. Admin tenant sees all sites; operator tenants see only their own.
- **Write (admin tenant only)**: site manifest changes require admin role. Audited to the feedback log (Section 10).

Sequence diagram



Reference implementation

- **Existing seed sites** in `src/sites.js` and `src/sites_energinet.js` — will be migrated to `sites/*.yaml` in a follow-up refactor.
- **Site loader:** `src/site_loader.js` (planned) — YAML parser + JSON schema validator + registry hookup.
- **Template:** `docs/sites/_template.md` — new-site flow doc scaffold.

Open questions

- **Multi-site tenants:** does one operator tenant own multiple sites (e.g. CPH Airports A/S owns CPH + Roskilde)? Current design says yes — `tenant` field can repeat across manifests.
- **Site-of-site (sub-sites):** does an airport have sub-sites for terminal / runway / apron? Current design says no — treat as one site with sensor placement metadata distinguishing zones.
- **Dynamic re-registration:** can a site's sensor list change without a boot? Current design says restart required. Consider hot-reload for large fleets.

4. Post-Incident Report export

Purpose

Every closed event produces a Post-Incident Report (PIR) as a structured record. This contract specifies the export format when a receiver / operator / auditor needs to bundle one or more PIRs into a portable file (PDF for reading, JSON for machine consumption, CSV for spreadsheet analysis).

Actors

- **Caller:** authorised viewer in the platform UI OR API client with `pir:export` scope
- **Receiver:** PIR export service (Azure Function, per-tenant instance)

Wire format

GET `/v1/pir/{event_id}/export?format={json|pdf|csv}&scope={self|chain}`

Query parameters:

- `format` — `json` (structured record), `pdf` (rendered report), `csv` (flattened for spreadsheet analysis)
- `scope` — `self` (only this event) or `chain` (this event + every event in the same xlink chain per Section 7 of cross-agency-flows.md)

Response headers:

```
Content-Type: application/json | application/pdf | text/csv
Content-Disposition: attachment; filename="PIR-{event_id}.{ext}"
X-Redaction-Applied: full | summary | mixed
X-Chain-Event-Count: 4
```

JSON export shape

```
{
  "export_version": 1,
  "generated_at": "2026-09-13T10:45:00.000Z",
  "generated_by": "receiver-tenant:politi-kbh",
  "correlation_id": "550e8400-e29b-41d4-a716-446655440000",
  "scope": "chain",
  "chain": {
    "chain_id": "chain-evt-1201",
    "size": 4,
    "first_at": "2026-09-13T09:12:00.000Z",
    "last_at": "2026-09-13T10:22:00.000Z",
    "span_minutes": 70,
    "sites": ["BILLUND", "CPH", "AALBORG"]
  },
  "events": [
    {
      "event_snapshot": {
        "id": "evt-1201",
        "site_id": "billund",
        "classification": "hostile",
        "threat": "high",
        "drone_type": "Shahed-136",
        "platform": "loitering-munition",
        "confidence": 0.94,
        "start_time": "2026-09-13T09:12:00.000Z",
        "end_time": "2026-09-13T09:47:00.000Z",
        "duration_sec": 2100,
        "outcome": "closed",
        "domain_scope": ["aviation", "ground"],
        "linked_event_ids": ["evt-1202", "evt-1203"]
      },
      "summary": "Shahed-136 signature detected at Billund airport perimeter..."
    }
  ]
}
```

```

"recommendation": "Trafikstyrelsen NOTAM restriction advised for next 30 min.",
"chapters": [
  {
    "role_id": "politi-kbh",
    "role_label": "Politi København",
    "primary_archetype": "kinetic-response",
    "visibility_level_for_viewer": "full",
    "involvement": {
      "cascades_in": 1,
      "cascades_out": 0,
      "responses_sent": 1,
      "dispatches_owned": 2,
      "dispatches_open": 0,
      "catalog_entries": 3
    },
    "populated_archetypes": ["kinetic-response", "coordination-command", "public-safety-communication"],
    "timeline": [ /* ... per-role timeline slice ... */ ],
    "sub_sections": { /* ... per-archetype sub-section content ... */ }
  }
  /* ... one chapter per contributor ... */
],
"master_timeline": [ /* ... unified event timeline ... */ ],
"escalations": [ /* ... per Section 6a of cross-agency-flows.md ... */ ],
"dispatches": [ /* ... per Section 6a ... */ ],
"catalog": {
  "subjects": [], "recordings": [], "resp_history": [],
  "attribution": [], "patterns": [], "xlinks": [], "roe": [],
  "evidence": [], "coord_decisions": [], "casualties": [],
  "advisories": [], "public_alerts": [], "liaison": []
}
}
/* ... additional events in the chain when scope=chain ... */
],
"redactions_applied": [
  {
    "role_id": "pet",
    "reason": "INTEL chapter compartmented from viewer archetype 'kinetic-response'",
    "level": "summary"
  }
]
}

```

PDF export

Rendered from the JSON via a server-side templating engine. Each event becomes one section; each contributor chapter becomes a sub-section. Chain scope produces one PDF with an events index up front. Redacted chapters render as "Redacted for tenant boundary — contact authoring branch for access."

CSV export

Flattened one-row-per-detection-event view. Useful for spreadsheet analysis (average response time per receiver, dispatch outcome distributions, etc). Loses the per-contributor chapter nesting; use JSON if you need the full structure.

Auth model

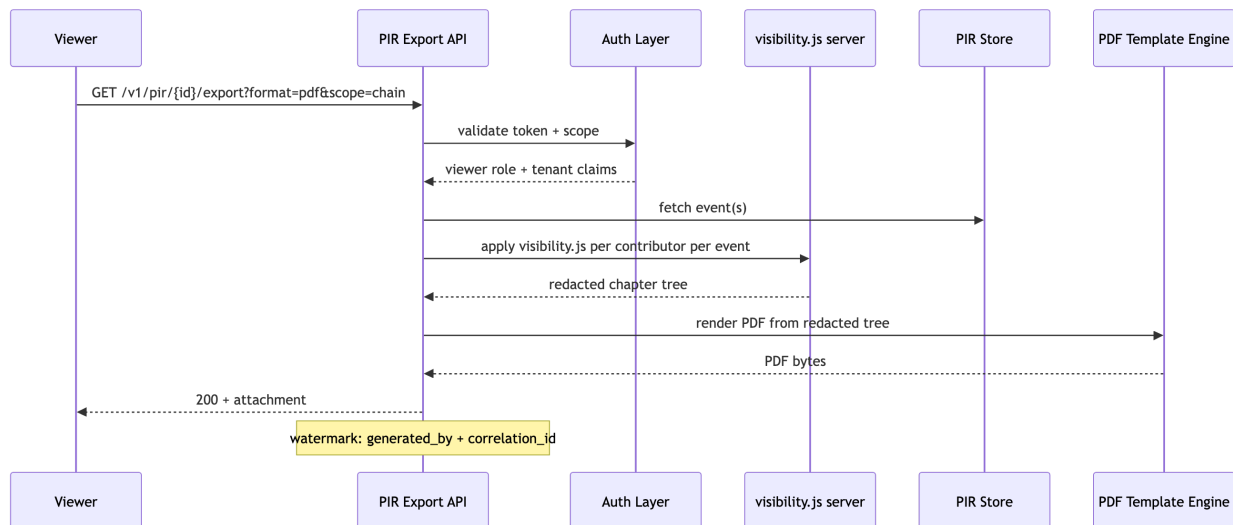
- OAuth 2.0 token with `pir:export` scope.

- Viewer's tenant + role claims determine which chapters render at what visibility level. Server-side `visibility.js` gate (Section 7) applies before export streams.
- Exports are watermarked with `generated_by` + `correlation_id` for audit traceability.

Error semantics

- **404 Not Found** — event id doesn't exist OR viewer isn't cleared to see it at all (HIDDEN visibility for every chapter of the event).
- **403 Forbidden** — viewer lacks `pir:export` scope.
- **410 Gone** — event closed >6 months ago and archival tier is offline. Retrieval requires ops ticket.

Sequence diagram



Reference implementation

- **Client-side JSON download:** already exists — `data-rcv="pir-download"` button in the Step 7 panel. Serves the local `event.postIncidentReport` object.
- **Server-side export:** planned as Azure Function. Reads from Azure Blob PIR archive, applies visibility, streams response.

Open questions

- **Chain scope + partial visibility:** if the chain has 4 events but the viewer is cleared to see only 2, should the export drop the other 2 entirely, or include their stubs with "no visibility" markers? Current design: drop entirely + a footer noting "N additional events in this chain were not exported due to visibility policy."
 - **Retention:** archival tier after 6 months? Legal + regulatory requirements TBD per customer.
 - **Cross-tenant export:** can Politi Kbh export a chain that includes an event from Esbjerg Port? Only if the operator tenant explicitly grants cross-tenant read via the Auth+IAM contract (Section 12).
-